

Hong Liu

Ph.D. student in Epidemiology and Health Statistics, School of Public Health, Xiamen University (admitted 2026, supervised by Prof. Tianmu Chen); M.Sc. in Computer Science and Technology, Macau University of Science and Technology; undergraduate training at Xiamen University; interdisciplinary background spanning public health, epidemiology, AI, and data science

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Profile

Academic training and current direction

I am a Ph.D. student in Epidemiology and Health Statistics at the School of Public Health, Xiamen University (admitted 2026, supervised by Prof. Tianmu Chen), and I hold an M.Sc. in Computer Science and Technology from Macau University of Science and Technology. My research focuses on infectious disease forecasting, influenza co-circulation analysis, outbreak modelling, and related research software.

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Education

Degrees and thesis work

Xiamen University

Ph.D. student in Epidemiology and Health Statistics (2026–present)

School of Public Health · doctoral study under Prof. Tianmu Chen

Doctoral research extends toward multimodal infectious disease spatiotemporal dynamics, viral evolution early warning, and integrated macro-micro modelling frameworks.

Macau University of Science and Technology

M.Sc. in Computer Science and Technology, Faculty of Innovation Engineering

Graduated · GPA 3.63/4 · Full scholarship and living allowance

Completed M.Sc. training in Computer Science and Technology under Academician Nanshan Zhong and Prof. Chitin Hon, with co-supervision by Prof. Tianmu Chen (Xiamen University). Graduate training emphasized intelligent control technology, artificial intelligence, numerical analysis, statistics, pattern recognition, algorithm theory, and optimization methods.

Xiamen University

B.Med. in Preventive Medicine (School of Public Health) & B.Econ. in Statistics (Wang Yanan Institute for Studies in Economics)

Preventive Medicine supervised by Prof. Tianmu Chen

Academic and project experience during the undergraduate stage laid the groundwork for infectious-disease modelling, health data analysis, and later AI-assisted research workflows. Selected high-mark coursework: Epidemiology, Biostatistics, and Health Statistics.

Master' s thesis

Multi-source-data-driven study of influenza spatiotemporal transmission dynamics, viral evolution, and early warning in China

Completed master's thesis direction

The completed thesis linked surveillance, environmental, behavioral, and sequence-related evidence while keeping completed work, submitted manuscripts, and future doctoral stages explicitly separated.

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Research Interests

Topics under active study

— Multimodal infectious disease forecasting and early warning.

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- Influenza co-circulation analysis, co-infection signals, and non-stationary time-series modelling.
 - Spatiotemporal transmission dynamics, small-sample outbreak modelling, and viral evolution.
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Publications and Manuscripts

Grouped by status, most recent first

Published papers 6

1. Spatiotemporal instability of influenza seasonality during viral co-circulation

Published

npj Systems Biology and Applications · 2026 · First author

First-author npj Systems Biology and Applications article. Accepted on 17 April 2026 and published online on 04 May 2026.

DOI: [10.1038/s41540-026-00729-9](https://doi.org/10.1038/s41540-026-00729-9)

2. Temporal shifts in 24 notifiable infectious diseases in China before and during the COVID-19 pandemic

Published

Nature Communications · 2024 · Co-author (7/14)

Co-authored publication listed in the source materials.

DOI: [10.1038/s41467-024-48201-8](https://doi.org/10.1038/s41467-024-48201-8)

3. Transmissibility of the variant of concern for SARS-CoV-2 in six regions

Published

Heliyon · 2024 · Co-author

Co-authored publication listed in the source materials.

DOI: [10.1016/j.heliyon.2024.e32164](https://doi.org/10.1016/j.heliyon.2024.e32164)

4. **Impact of long-term care insurance on the physical and mental health of older adults in China**

Published

Medicine and Society · 2024 · Co-author (2/5)

Co-authored publication listed in the source materials.

DOI: 10.13723/j.yxysh.2024.08.011

5. **Assessment of transmissibility and measures effectiveness of SARS in 8 regions, China,2002-2003**

Published

Frontiers in Cellular and Infection Microbiology · 2023 · Co-author (4/13)

Co-authored publication listed in the source materials.

DOI: 10.3389/fcimb.2023.1212473

6. **Advancements in Defining and Estimating the Reproduction Number in Infectious Disease Epidemiology**

Published

China CDC Weekly · 2023 · Co-author (7/11)

Co-authored publication listed in the source materials.

DOI: 10.46234/ccdcw2023.158

Preprints 2

1. **Multi-modal Adaptive Estimation for Temporal Respiratory Disease Outbreak**

Preprint

arXiv preprint · 2025 · First author

Status: Public preprint; journal submission under review

First-author MAESTRO study. In the documented evaluation context, the reported R^2 reached 0.956 on a 10-year Hong Kong influenza dataset.

DOI: 10.48550/arXiv.2509.08578

2. **Three machine-learning models to predict dengue vectors Breteau Index based on meteorology and biotope in Fujian China**

Preprint

Research Square (preprint) · 2024 · Co-author (2/10)

Status: Preprint

Preprint posted on Research Square in April 2024; Ke is the first author and Hong Liu is a contributing co-author (2/10). No published journal version has been identified.

DOL: 10.21203/rs.3.rs-4286413/v1

Submitted / under review 2

1. **Dual-Model Framework for CHIKV Transmission Modeling: ODE and Petri Net Analysis of the 2025 Foshan Outbreak**

Under review

Epidemics · 2025 · First author

Status: Submitted; with editor

First-author dual-framework outbreak modelling study for the 2025 Foshan chikungunya outbreak. A publicly available arXiv preprint is openly accessible.

DOL: 10.48550/arXiv.2512.12577 Manuscript ID: EPIDEMICS-D-25-00174

2. **Psychological resilience, occupational stress, organizational trust, and burnout among healthcare workers**

Under review

Health Soft Science · 2025 · Co-author (2/6)

Status: Submitted; with editorial office

Questionnaire-based collaborative study using CFA, SEM, and bootstrap analysis.

Manuscript ID: 251802

Research Software and Projects

Code, tooling, and data platforms

MultiCoPat Public repository

Multi-region and multi-window influenza co-circulation analysis and prediction workflows.

[Influenza](#)[Co-circulation](#)[Prediction](#)[Time series](#)

github.com/Hylouis233/MultiCoPat

MAESTRO Public repository

Multimodal forecasting framework for respiratory disease activity and early-warning-oriented time-series analysis.

[Forecasting](#)[Early warning](#)[Multimodal modelling](#)[Time series](#)

github.com/Hylouis233/MAESTRO

ODE-Petri-Chikungunya Public repository

Research workflow comparing ODE and Petri Net perspectives for the 2025 Foshan chikungunya outbreak.

[Mechanistic modelling](#)[ODE](#)[Petri Net](#)[Outbreak analysis](#)

github.com/Hylouis233/ODE-Petri-Chikungunya

Breteau Index Prediction Model Public repository

Supplementary codebase for machine learning-based dengue vector Breteau Index prediction.

[Machine Learning](#)[Prediction Model](#)[Breteau Index](#)

github.com/Hylouis233/Breteau-Index-Prediction-Model-using-machine-learning

Infectious disease intelligence platform

Mixed public/private codebases; selected components are private

Crawler, database, ETL, and visualization workflow for infectious-disease news collection, monitoring, and research-oriented data support.

ETL

Crawler

Database

Visualization

No public repository link is shown for this item.

bibverify Referenced public repository

BibTeX validation and metadata completion tool for DOI, author, year, deduplication, and citation normalization workflows.

Bibliographic tooling

DOI validation

Metadata completion

github.com/Hylouis233/bibverify

MCM 2023 Problem C Solution Public repository

Team solution and codebase for the Mathematical Contest in Modeling (MCM) 2023 Problem C.

Mathematical Modeling

MCM

Data Analysis

github.com/Hylouis233/MCM2023C

Baidu Index Spider Public repository

A Baidu Index scraping tool equipped with a graphical user interface (GUI).

Spider

Data Collection

GUI

github.com/Hylouis233/baidu-index-spider

GMCM LaTeX Template Public repository

Refactored LaTeX templates for the China Graduate Mathematical Contest in Modeling.

LaTeX

Template

Mathematical Modeling

github.com/Hylouis233/GMCM_LaTeX

Awards

Selected honors and competition results

2025	Regional First Prize, 15th MathorCup Mathematical Application Challenge (Graduate Division) China Society for Overall Planning, Optimization and Economic Mathematics
2025	National Third Prize, 15th MathorCup Mathematical Application Challenge (Graduate Division) China Society for Overall Planning, Optimization and Economic Mathematics
2025	Second Prize, 20th Citibank Financial Innovation Application Competition Shanghai University Student Science and Technology Entrepreneurship Foundation; Citigroup Financial Information Services (China) Co., Ltd.
2024	Third Prize, 21st China Graduate Mathematical Contest in Modeling (“Huawei Cup”) China Graduate Mathematical Contest in Modeling Organizing Committee
2024	Project Innovation Award, 13th “Win in Guangzhou” Hong Kong–Macao Division “Win in Guangzhou” University Entrepreneurship Competition Organizing Committee

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Experience

Research, coursework, and service

2024–2025

Macau University of Science and Technology

Completed M.Sc. coursework

Completed coursework in intelligent control technology, artificial intelligence, numerical analysis, statistics, pattern recognition, algorithm theory, and optimization methods, with a cumulative GPA of 3.63/4.

2019–2021

Xiamen University

Student service roles

Served as a student staff member in the medical school organization department and later as deputy head in the innovation service division of the student technology, entrepreneurship, and employment service center.

2024–2025

Collaborative laboratory project

Laboratory escape-risk digital twin project

Participated in full-stack development for a P3 laboratory digital twin and risk-assessment workflow involving Unity3D, Python, Flask, MySQL, and scenario simulation.

2024–2025

Collaborative laboratory project

Protein design and structure-biology tooling support

Supported target-site discovery and primer design workflows using masif_seed, Rosetta, AlphaFold 3, and RFDiffusion.

2024–present

Independent project

Infectious-disease news data platform

Independently developed and maintained a platform covering data collection, cleaning, ETL, database ingestion, and visualization for infectious-disease news and event monitoring.

2024–2025

Collaborative health research project

Questionnaire data cleaning and SEM analysis

Handled large-scale questionnaire cleaning, CFA, SEM, reliability and validity analysis, and bootstrap-based inference for healthcare workforce research.

2020–2024

Xiamen University and related teams

Selected practice and volunteer activities

Participated in outreach, practice, and volunteer activities including recruitment outreach, public-health themed practice teams, and report writing.

2024–2025

Modelling and innovation competitions

Competition projects

Contributed modelling, optimization, software implementation, and evaluation work in the Huawei Cup, MathorCup, Citibank competition, and related innovation contests.